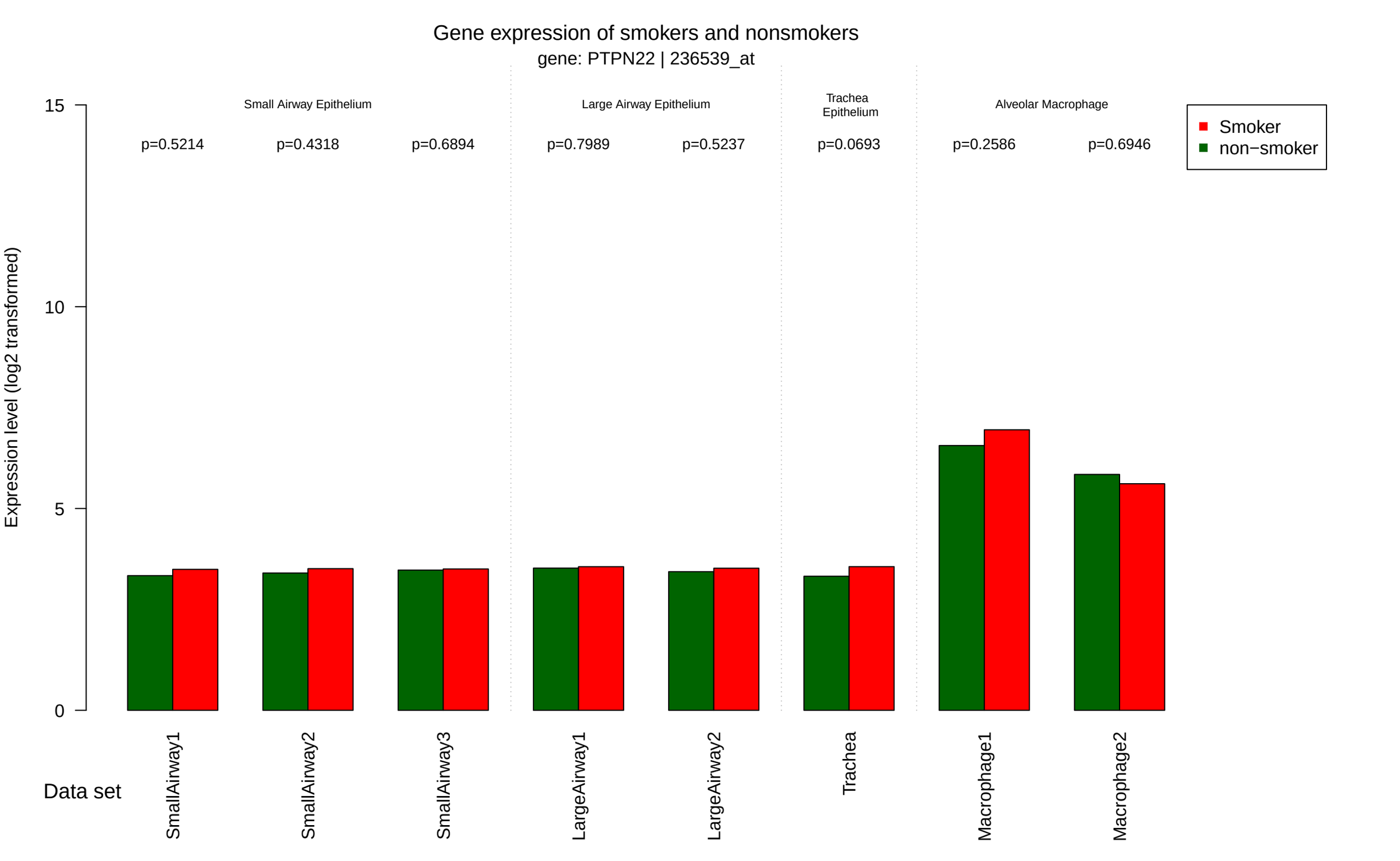
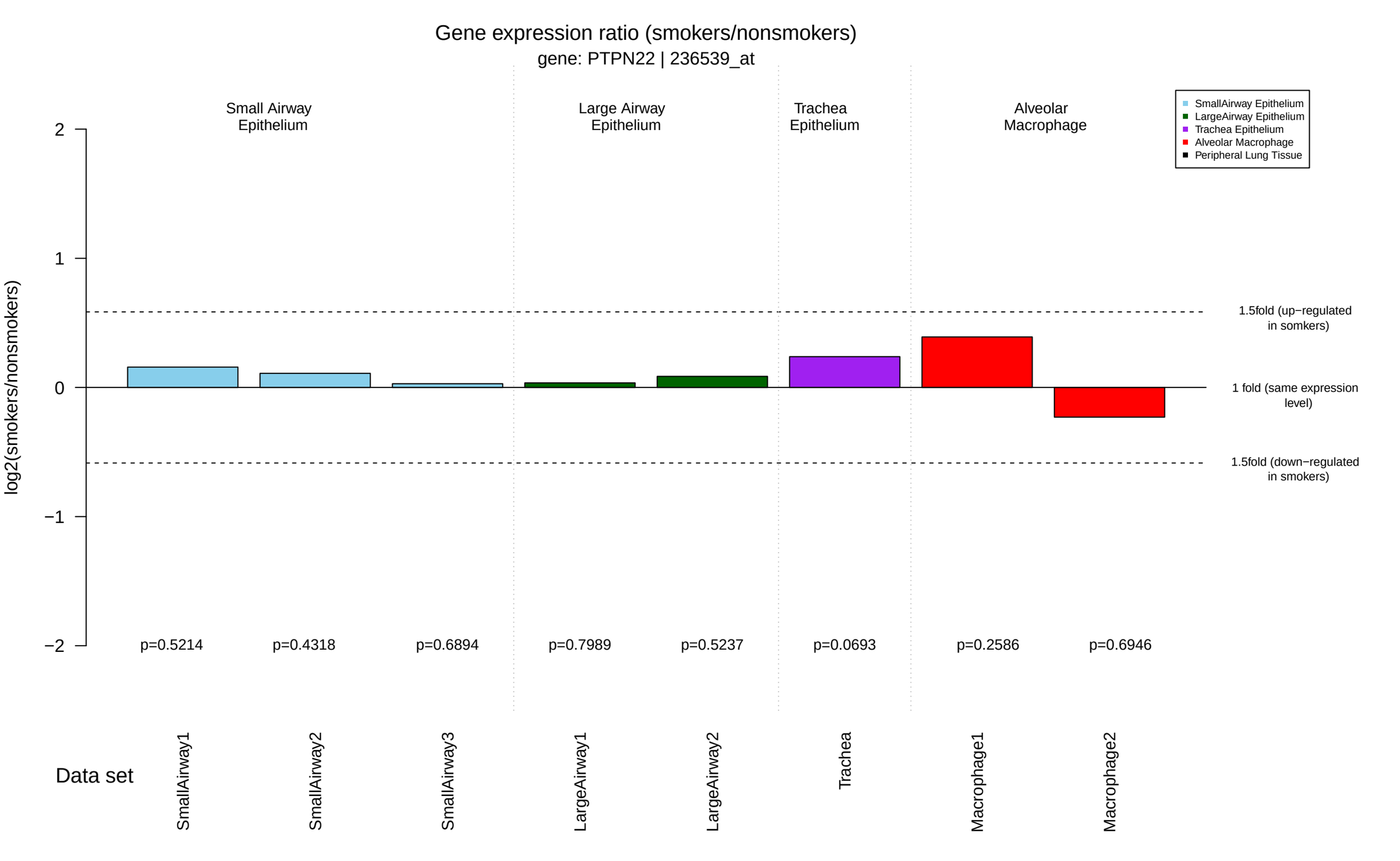
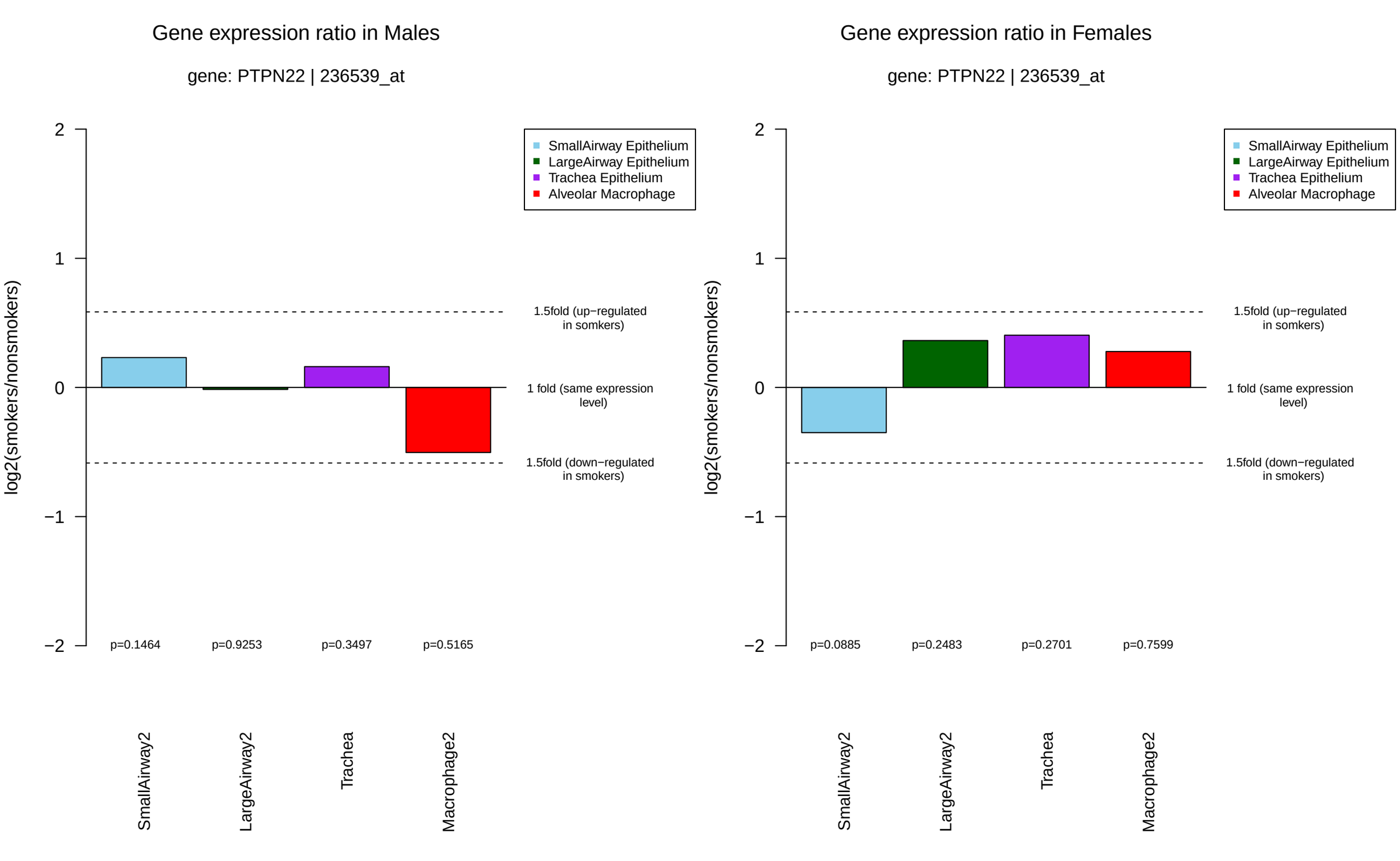
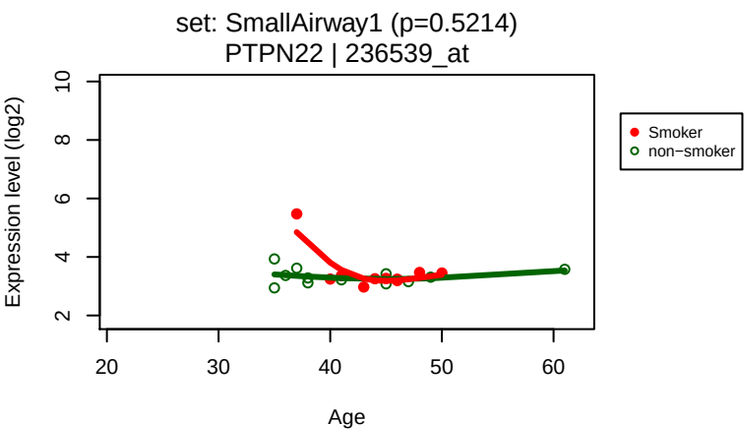
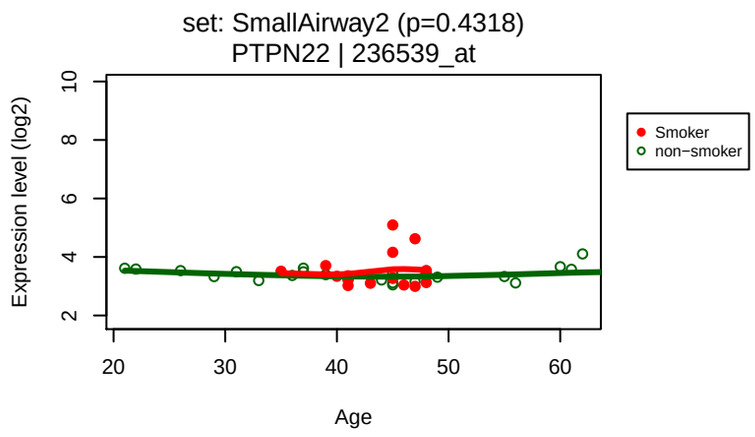
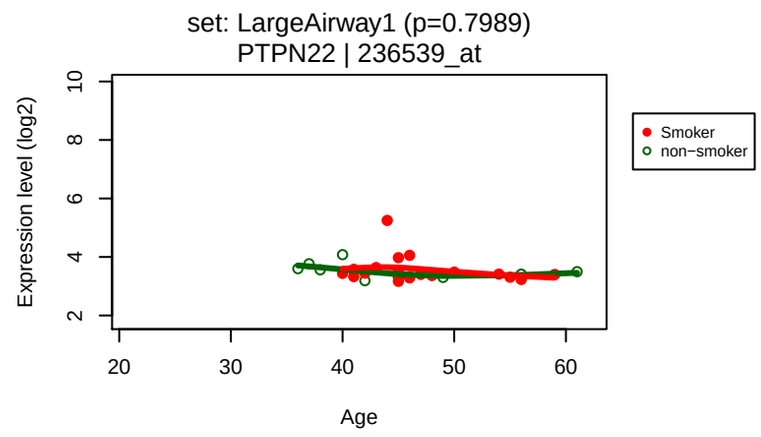

Gene expression of smokers and nonsmokers
gene: PTPN22 | 236539_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5214
p=0.4318
p=0.6894
p=0.7989
p=0.5237
p=0.0693
p=0.2586
p=0.6946
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

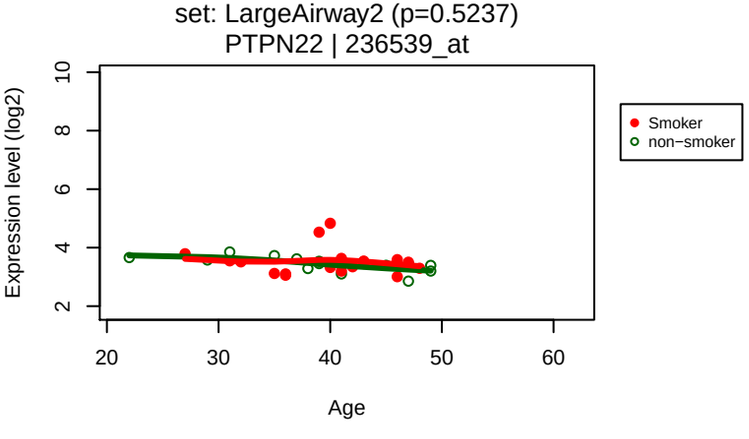

Gene expression ratio (smokers/nonsmokers)
gene: PTPN22 | 236539_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5214
p=0.4318
p=0.6894
p=0.7989
p=0.5237
p=0.0693
p=0.2586
p=0.6946
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

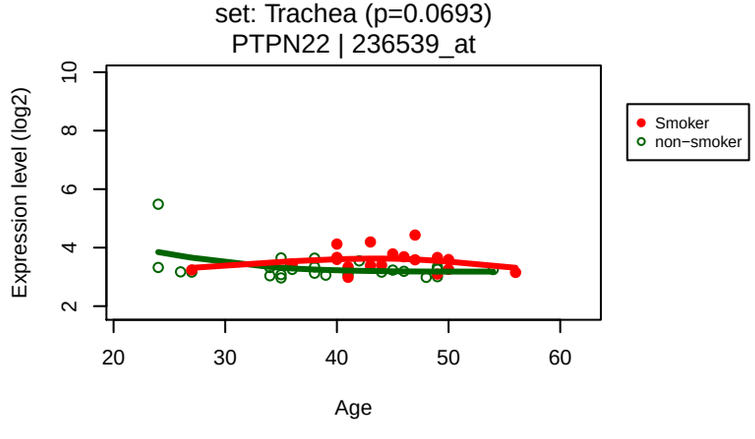

Gene expression ratio in Males
gene: PTPN22 | 236539_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1464
p=0.9253
p=0.3497
p=0.5165
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PTPN22 | 236539_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0885
p=0.2483
p=0.2701
p=0.7599
SmallAirway2
LargeAirway2
Trachea
Macrophage2

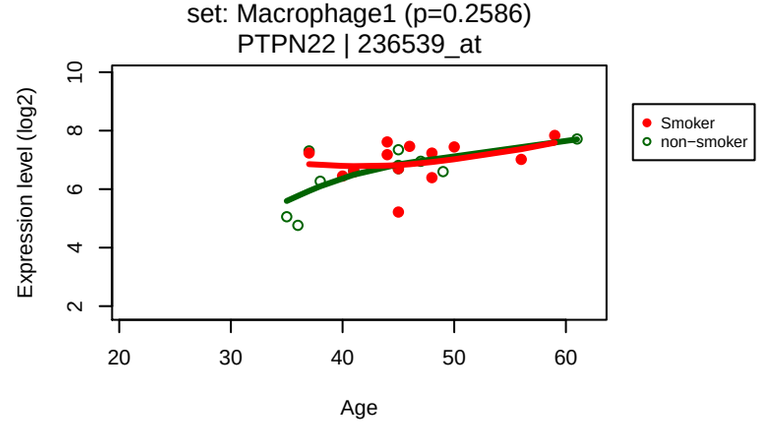

set: SmallAirway1 (p=0.5214)
PTPN22 | 236539_at
Expression level (log2)
Age
Smoker
non−smoker

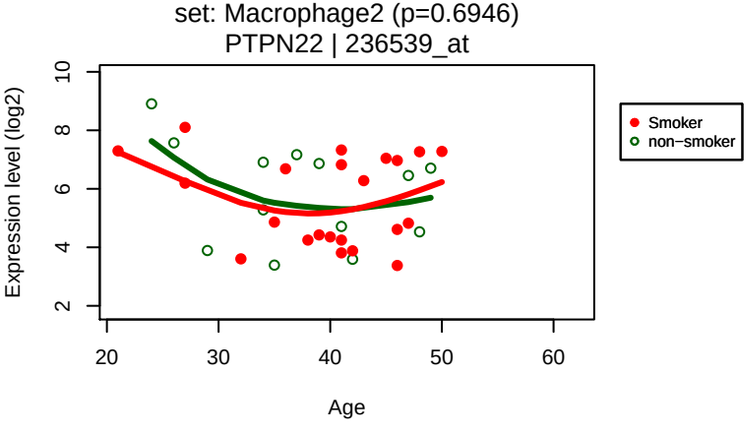

set: SmallAirway2 (p=0.4318)
PTPN22 | 236539_at
Expression level (log2)
Age
Smoker
non−smoker

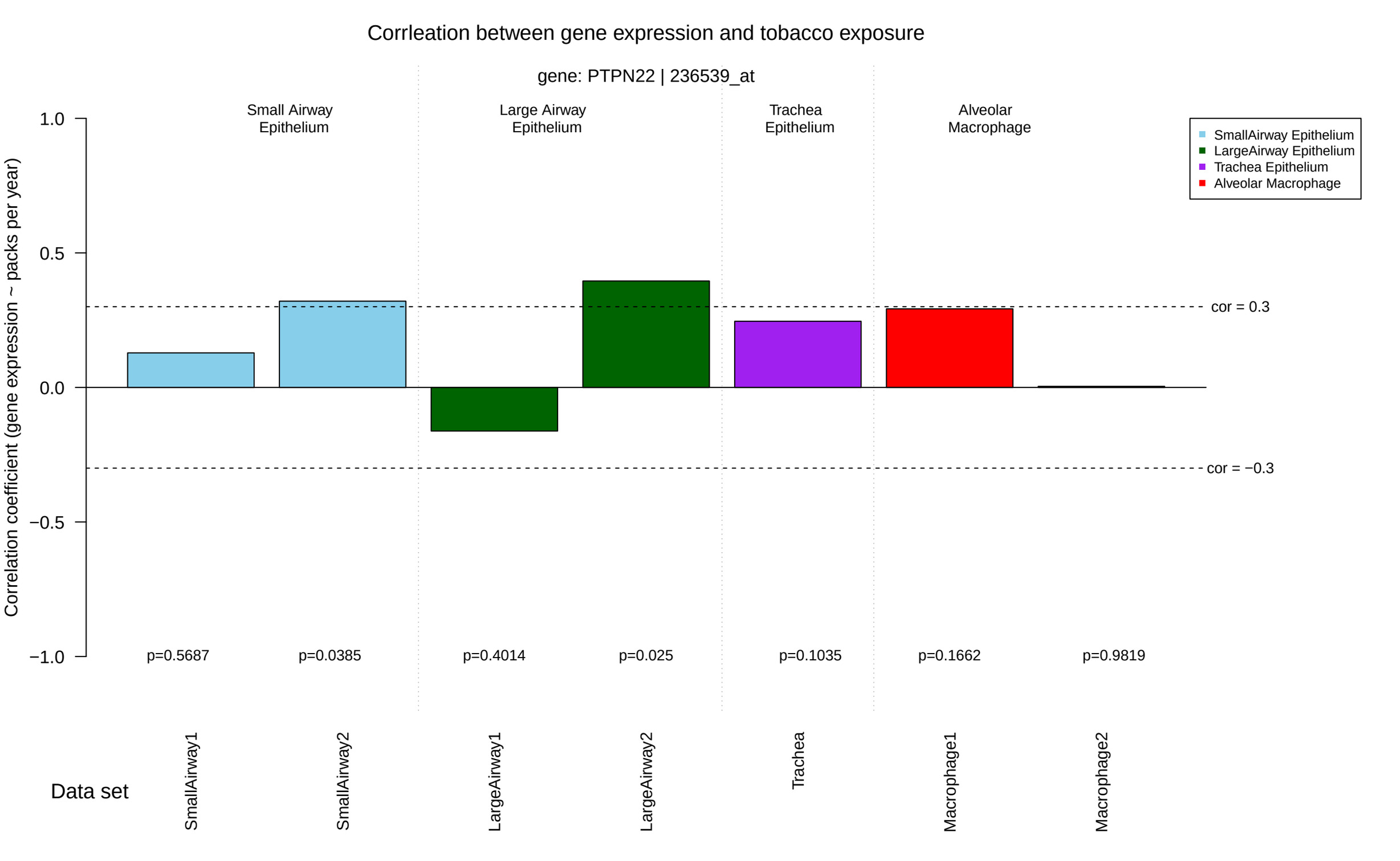

set: LargeAirway1 (p=0.7989)
PTPN22 | 236539_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5237)
PTPN22 | 236539_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0693)
PTPN22 | 236539_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2586)
PTPN22 | 236539_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6946)
PTPN22 | 236539_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PTPN22 | 236539_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5687
p=0.0385
p=0.4014
p=0.025
p=0.1035
p=0.1662
p=0.9819
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2